Pearson Edexcel Level 3 GCE

Friday 15 May 2026

Afternoon

Paper reference **8FM0/27**

Further Mathematics

Advanced Subsidiary
Further Mathematics options
27: Decision Mathematics 1
(Part of options D, F, H and K)

You must have:
Mathematical Formulae and Statistical Tables (Green), calculator,
D1 Answer Book (enclosed)

Candidates may use any calculator allowed by Pearson regulations. Calculators must not have the facility for symbolic algebra manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

Instructions

- Use **black** ink or ball-point pen.
- If pencil is used for diagrams/sketches/graphs it must be dark (HB or B).
- **Fill in the boxes** at the top of the answer book with your name, centre number and candidate number.
- Answer **all** questions and ensure that your answers to parts of questions are clearly labelled.
- Answer the questions in the answer book provided
  – *there may be more space than you need*.
- You should show sufficient working to make your methods clear. Answers without working may not gain full credit.
- Inexact answers should be given to three significant figures unless otherwise stated.
- Do not return the question paper with the D1 Answer Book.

Information

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- The total mark for this part of the examination is 40. There are 5 questions.
- The marks for **each** question are shown in brackets
  – *use this as a guide as to how much time to spend on each question*.

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

Turn over ▶

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1.

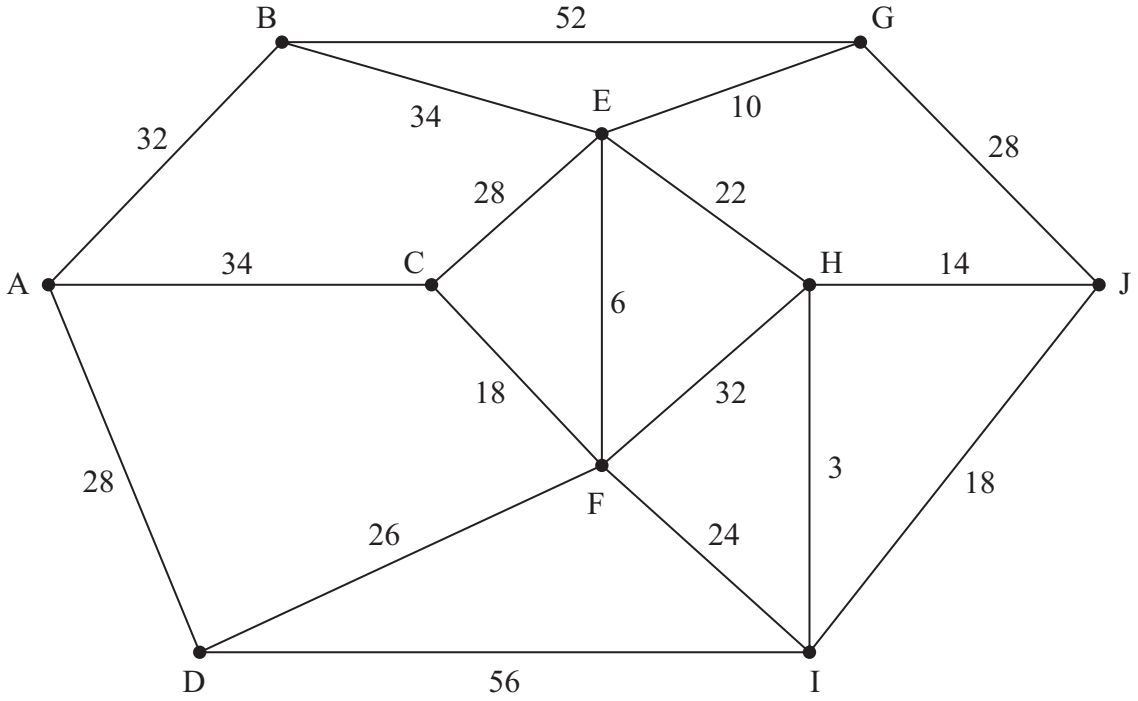


Figure 1

(i) Figure 1 represents a network of cycle tracks. The number on each arc represents the time taken, in minutes, to cycle along the corresponding track.

(a) Use Dijkstra's algorithm to find the shortest time needed to cycle from A to J.

(b) State the quickest route from A to J.

(6)

(ii) One application of Dijkstra's algorithm has order n^2 where n is the number of nodes in the network.

It takes a computer 0.32 seconds to find the shortest path from a given start node to a given end node in a network of 20 nodes.

Calculate approximately how long it would take, in minutes, for the computer to find the shortest path from a given start node to a given end node for a network of 4500 nodes.

(2)

(Total for Question 1 is 8 marks)

2.

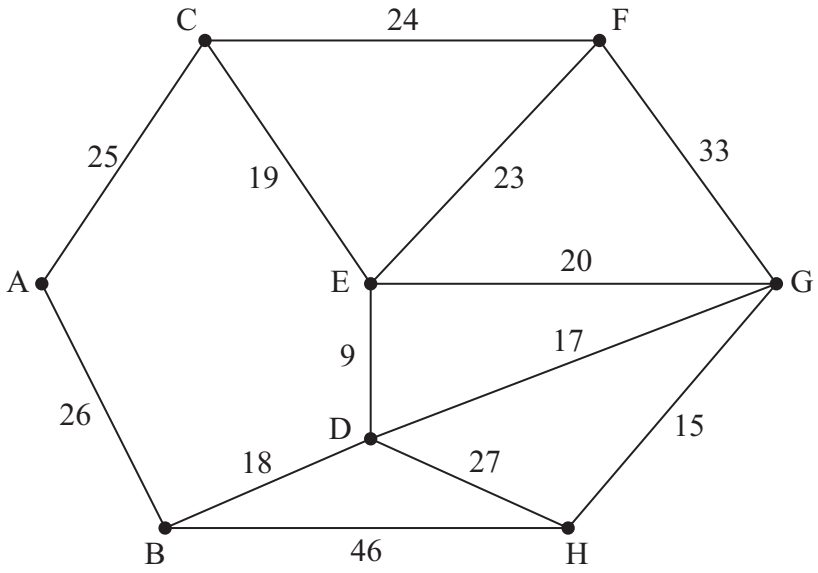


Figure 2

[The total weight of the network is 302]

Roads in a network connecting eight towns, A, B, C, D, E, F, G and H, are represented in Figure 2. The number on each arc represents the length, in miles, of the corresponding road.

(i) (a) Use Kruskal's algorithm to find the minimum spanning tree for the network. You should list the arcs in the order in which you consider them. In each case, state whether or not you are adding the arc to your minimum spanning tree.

(3)

(b) State the weight of the minimum spanning tree.

(1)

(ii) A route is needed that travels along each road at least once. The route must start and finish at A and the length of the route should be minimised.

(a) By considering the pairings of all relevant nodes, find the roads that need to be traversed twice.

(4)

(b) State the total length of this route.

(1)

(Total for Question 2 is 9 marks)

3.

32 45 17 23 38 28 16 9 12 10

The list of ten numbers above is to be sorted into **descending** order.

(a) Perform a bubble sort to obtain the sorted list, giving the state of the list after each completed pass.

(3)

The ten numbers are to be packed into bins of size n, where n is a positive integer.

When the first-fit bin packing algorithm is applied to the **original** list of ten numbers, the following allocation is obtained.

Bin 1:	32	17	9
Bin 2:	45	12	
Bin 3:	23	28	
Bin 4:	38	16	
Bin 5:	10		

(b) Determine the possible values of n, justifying your answer.

(3)

(Total for Question 3 is 6 marks)

4.

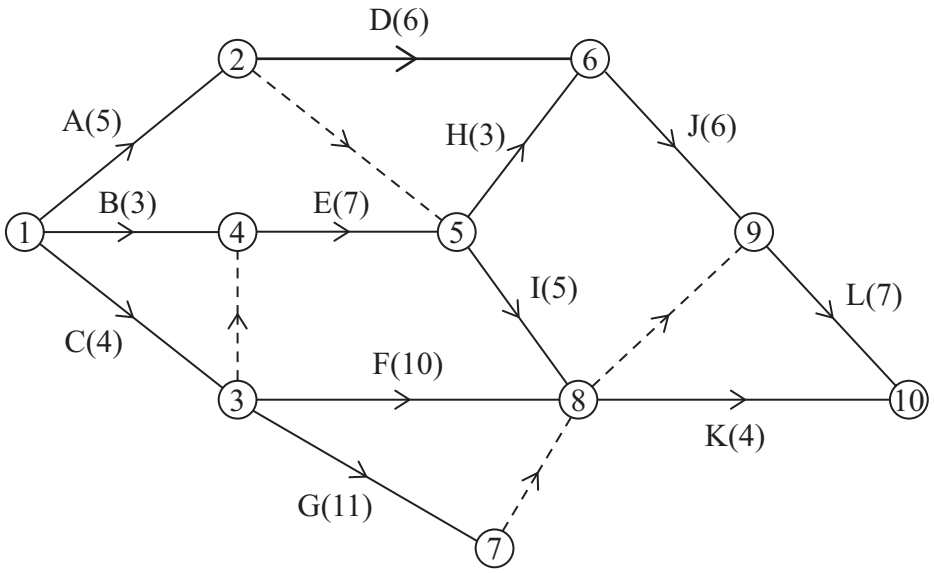


Figure 3

A project is modelled by the activity network shown in Figure 3. The activities are represented by the arcs. The number in brackets on each arc gives the time required, in hours, to complete the corresponding activity. The numbers in the circles are the event numbers.

(a) Complete the table in the answer book to show the immediately preceding activities for activities H, I, J, K and L.

(2)

(b) Complete the diagram in the answer book to show the early event times and the late event times.

(3)

(c) State the critical activities.

(1)

(d) Draw a Gantt (cascade) chart for this project on the grid in the answer book.

(4)

(Total for Question 4 is 10 marks)

5. A cafe owner plans to buy coffee cups for their cafe. The three sizes of cup they can buy are small, medium and large.

- small cups cost £3 each, medium cups cost £4 each and large cups cost £5 each
- the cafe owner decides to buy 200 cups in total
- they decide to buy at most 5 medium cups for every 3 large cups that they buy
- the number of large cups they buy must be at most 35% of the total number of small and medium cups combined

The cafe owner wishes to minimise the total cost of the cups that they buy.

Let x represent the number of small cups, y represent the number of medium cups and z represent the number of large cups that the cafe owner will buy.

Formulate this as a linear programming problem **in x and y only**, clearly showing your working. You must state the objective and list the constraints as simplified inequalities with integer coefficients.

You should **not** attempt to solve the problem.

(Total for Question 5 is 7 marks)

TOTAL FOR DECISION MATHEMATICS 1 IS 40 MARKS

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